

1 Online task and resource signals

Local observations

Coordinates, targets, tasks status ...

Battery levels

Current levels, changes

Tasks events

Delivered, picked-up, charge, package requests ...

Agent role



Package handling events



2 Relationship classification

AGVs



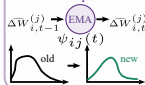
Pickers



Symbiosis classes

— Mutualism
- - - Commensalism
... Neutral

EMA update



3 PRISM reward decomposition

Rewards

$$r_i(t) = r_i^{\text{task}}(t) + r_i^{\text{sym}}(t)$$

Task reward

Symbiotic reward

Pair-potential shaping

$$\Phi_i \leftarrow \sum_j \varphi(\text{rel}_{ij}, t), \delta_{ij}, t$$

Typed φ (relationship weight)

φ^{mut} φ^{com} φ^{neu}

Potential update

$$r_i^{\text{sym}}(t) \leftarrow \beta[\gamma \Phi_i(t+1) - \Phi_i(t)]$$

4 Policy optimization

Shaped reward

$$\{r_i^{\text{task}} + r_i^{\text{sym}}\}$$

IPPO/MAPPO

AGV policy π_{θ}^A



Actions
Move
Pack
Drop
Charge
Wait ...

Centralized critic shared by type

Picker policy π_{θ}^P



Actions
Move
Pick up
Drop
Charge
Wait ...

Execute decentralized actions (move/pickup/pack/drop/charge/wait...)